The Consequences of Not Optimally-Stopping Agriculture

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Abstract

Agricultural production is commonly discussed in terms of maximizing output, productivity, yield, and profitability. This paper examines the complementary proposition that agricultural production should also be stopped at an economically appropriate point. Building on the algebraic framework introduced in "Agricultural economics" [1], the paper defines yield, dividend, and reap through the sequential transformation from seeds to plants to fruits. The resulting fundamental equation is

$$r = y + d + yd. \tag{1}$$

The corresponding inverse representation is

$$y = \frac{r-d}{1+d}. \tag{2}$$

The paper's proposed stopping condition is

$$1 + y = \frac{1+r}{1+d}. \tag{4}$$

The central argument developed here is that the stopping condition is potentially valuable because its constituent quantities are constructed from observable agricultural quantities and can therefore be estimated using modern statistical and artificial-intelligence methods. Failure to stop appropriately can, under competitive or imperfectly competitive market conditions, contribute to excess supply, price pressure, revenue deterioration, and strategic undercutting. The paper develops these implications through agricultural economics, finance, biology, chemistry, physics, statistics, machine learning, behavioral science, welfare economics, political economy, geopolitics, and warfare economics. The interdisciplinary extensions are explicitly distinguished from the mathematical content of the original paper.

The paper ends with "The End"

1 Introduction

The conventional agricultural objective is frequently expressed as increasing production. More seeds may generate more plants, and more plants may generate more fruits. However, production is not conducted in a vacuum. Agricultural output enters a market in which consumers, competitors, intermediaries, processors, governments, and international markets jointly determine the economic value realized by the producer.

Consequently, a distinction must be made between physical production and economically optimal production.

The original paper "Agricultural economics" introduces a deliberately simple sequential production framework. Let the number of seeds be s, the number of plants be P, and the number of fruits be F. It defines yield, dividend, and reap from these quantities and derives a fundamental equation for the complete production process [1].

The present paper develops the economic consequences of the associated stopping condition. The principal proposition is not that agriculture should necessarily produce less. Rather, it is that agricultural production should be treated as a dynamic decision problem in which continuation has an opportunity cost.

2 The original agricultural production framework

The original paper defines the yield as

$$y = \frac{P}{s} - 1, \tag{1}$$

the dividend as

$$d = \frac{F}{P} - 1, \tag{2}$$

and the reap as

$$r = \frac{F}{s} - 1. \tag{3}$$

These definitions correspond to two sequential production transformations:

$$s \longrightarrow P \longrightarrow F. \tag{4}$$

The first transformation converts seeds into plants. The second converts plants into fruits. The complete transformation converts seeds into fruits.

The corresponding gross transformation factors are

$$1 + y = \frac{P}{s}, \tag{5}$$

$$1 + d = \frac{F}{P}, \tag{6}$$

and

$$1 + r = \frac{F}{s}. \tag{7}$$

The structure therefore implies

$$1 + r = \frac{F}{s} \qquad = \frac{F}{P}\frac{P}{s} \quad = (1 + d)(1 + y). \tag{8}$$

Expanding gives

$$1 + r = 1 + y + d + yd, \tag{9}$$

and hence

$$\boxed{r = y + d + yd}. \tag{1}$$

This is equation (1) of the original paper and is explicitly called there the “fundamental equation of agricultural economics” [1].

3 Inverse representation

Solving the fundamental equation for yield gives

$$r = y + d + yd, \; r - d \qquad = y(1 + d), \tag{10}$$

and therefore

$$\boxed{y = \frac{r - d}{1 + d}.} \tag{2}$$

This is equation (2) of the original paper [1].

The importance of this inversion is practical. Instead of forecasting the complete reap from the component quantities, one may infer the yield compatible with an observed reap and dividend.

4 The third equation and its algebraic structure

The original paper gives the following exact identity:

$$\boxed{\frac{r-d}{1+d} = r - d(1-d)(1+d^2)(1+r) - \frac{d^5(1+r)}{1+d}.} \tag{3}$$

This is equation (3) of the original paper [1].
The identity can be verified directly.
First,

$$d(1-d)(1+d^2) = d - d^2 + d^3 - d^4. \tag{11}$$

Furthermore,

$$d - d^2 + d^3 - d^4 + \frac{d^5}{1+d} = \frac{d}{1+d}. \tag{12}$$

Therefore,

$$\begin{aligned}
r - d(1-d)(1+d^2)(1+r) - \frac{d^5(1+r)}{1+d} & \qquad (13)\\
= r - (1+r)\left(d - d^2 + d^3 - d^4 + \frac{d^5}{1+d}\right) & \qquad (14)\\
= r - \frac{d(1+r)}{1+d} & \qquad (15)\\
= \frac{r(1+d) - d(1+r)}{1+d} & \qquad (16)\\
= \frac{r + rd - d - dr}{1+d} & \qquad (17)\\
= \frac{r-d}{1+d}. & \qquad (18)
\end{aligned}$$

Thus equation (3) is exactly equivalent to equation (2).
The third equation can also be understood through the finite geometric identity

$$\frac{1}{1+d} = 1 - d + d^2 - d^3 + d^4 - \frac{d^5}{1+d}. \tag{19}$$

Multiplying this identity by d gives

$$\frac{d}{1+d} = d - d^2 + d^3 - d^4 + \frac{d^5}{1+d}, \tag{20}$$

which is precisely the decomposition required for equation (3).

5 The proposed stopping condition

The original paper states that agriculture should be stopped optimally when

$$\boxed{1 + y = \frac{1+r}{1+d}.} \tag{4}$$

This follows immediately from equation (2):

$$\begin{aligned}
y &= \frac{r-d}{1+d}, & (21)\\
1 + y &= 1 + \frac{r-d}{1+d} & (22)\\
&= \frac{1+d+r-d}{1+d} & (23)\\
&= \frac{1+r}{1+d}. & (24)
\end{aligned}$$

The mathematical condition is therefore equivalent to the fundamental production identity.
The crucial economic question is why such a simple equality should have economic value.

6 Why stopping can be economically important

Physical production and economic value are not identical.

Suppose an agricultural producer continues production after the economically appropriate stopping point. The additional output must ultimately be absorbed by some combination of consumers, processors, traders, storage facilities, government procurement systems, exports, or competing markets.

If market demand does not expand sufficiently rapidly, additional supply can place downward pressure on prices. Agricultural market research recognizes the possibility that oversupply can generate price reductions and competitive responses among producers [2]. Agricultural policy literature likewise identifies the avoidance of burdensome surpluses and excessive price fluctuations as important objectives [4].

The relevant economic distinction is therefore

$$\text{maximum physical output} \neq \text{maximum economic value}. \tag{25}$$

A producer may be technically capable of producing another unit while being economically better off not producing it.

7 Low-balling and competitive undercutting

Let the market price be p and agricultural quantity supplied be Q. A simple representation is

$$p = p(Q), \tag{26}$$

with

$$\frac{dp}{dQ} < 0 \tag{27}$$

over a relevant downward-sloping portion of the market-demand relationship.

If a producer increases output while aggregate demand remains relatively fixed, market-clearing pressure can reduce price.

The revenue function is

$$R(Q) = p(Q)Q. \tag{28}$$

Its derivative is

$$\frac{dR}{dQ} = p(Q) + Q\frac{dp}{dQ}. \tag{29}$$

The second term is economically important. When additional quantity reduces the market price on existing units, the producer may incur an inframarginal price effect in addition to the revenue generated by the marginal unit.

In a competitive environment, one producer's additional supply can interact with the supply decisions of other producers. In an imperfectly competitive environment, producers may respond strategically.

This creates a potential sequence:

$$\begin{matrix}\text{continued}\\\text{production}\end{matrix} \longrightarrow \begin{matrix}\text{additional}\\\text{supply}\end{matrix} \longrightarrow \begin{matrix}\text{price}\\\text{pressure}\end{matrix} \longrightarrow \begin{matrix}\text{competitive}\\\text{undercutting}\end{matrix} \longrightarrow \begin{matrix}\text{lower}\\\text{producer.}\\\text{revenue}\end{matrix} \tag{30}$$

"Low-balling" is used here as an informal description of selling at unusually low prices in response to excess supply or competitive pressure. It is not a term introduced formally by the original paper.

8 A simple game-theoretic extension

Consider two producers, i and j. Let their quantities be q_i and q_j, respectively. Aggregate supply is

$$Q = q_i + q_j. \tag{31}$$

Suppose the inverse demand function is

$$p(Q) = a - bQ, \qquad a > 0, \quad b > 0. \tag{32}$$

Producer i's revenue is

$$R_i(q_i, q_j) = q_i\left(a - b(q_i + q_j)\right). \tag{33}$$

The corresponding marginal revenue is

$$\frac{\partial R_i}{\partial q_i} = a - 2bq_i - bq_j. \tag{34}$$

Thus producer i's optimal quantity depends not only on its own production but also on the quantity selected by its competitor.

The strategic implication is immediate: continuation of production can change the market environment faced by all producers.

This provides an economic interpretation of the stopping principle. The stopping decision should not be regarded solely as an agronomic decision. It can be a strategic market decision.

9 Finance and agricultural capital allocation

Agriculture is also an investment process. Seeds, fertilizer, land, labor, water, machinery, pesticides, energy, storage, and logistics constitute capital and operating inputs.

Let the incremental cost of additional production be MC and the expected incremental revenue be MR. A conventional marginal decision criterion is

$$MR - MC > 0. \tag{35}$$

Continuation is economically attractive when the expected marginal benefit exceeds the marginal cost. Conversely,

$$MR - MC < 0 \tag{36}$$

provides an economic reason to stop.

The stopping condition proposed in the original paper is not identical to this conventional marginal-profit condition. Rather, it supplies a production-side algebraic boundary that can be incorporated into a richer economic decision system.

This distinction is important: equation (1) is an agricultural transformation identity, whereas a complete profit-maximization model would additionally require prices, costs, constraints, uncertainty, and preferences.

10 Biology

The biological interpretation is particularly natural.

The sequence

$$s \longrightarrow P \longrightarrow F \tag{37}$$

represents a simplified life-cycle production chain.

Biologically, however, the transformation is constrained by finite resources. Let X denote available environmental resources. Then production can be represented abstractly as

$$P = f_1(s, X), \tag{38}$$

and

$$F = f_2(P, X). \tag{39}$$

The functions need not be linear.

Examples of biological constraints include:

- soil nutrient availability;
- water availability;
- sunlight;
- temperature;

- disease pressure;
- pest populations;
- plant density;
- pollination;
- genetic characteristics;
- physiological maturity.

The economic stopping problem therefore sits on top of a biological production system.

11 Chemistry

Agricultural production is also a chemical transformation system. Plant growth depends upon the availability and cycling of chemical elements, including carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, sulfur, and micronutrients.

Let the vector of relevant resources be

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)^\mathsf{T}. \tag{40}$$

A generalized production function may be written as

$$\mathbf{z} = f(\mathbf{x}, \boldsymbol{\theta}), \tag{41}$$

where $\boldsymbol{\theta}$ represents biological and environmental parameters.

Overuse of an input can have nonlinear effects. Consequently, increasing an agricultural input does not imply that output increases proportionally.

The original paper's multiplicative structure is therefore compatible with a broader scientific interpretation in which successive transformations interact rather than simply add.

12 Physics and threshold phenomena

The stopping problem has an analogous mathematical structure to threshold decisions in physical systems.

Let $x(t)$ represent a production state. A general dynamic system can be written

$$\frac{dx}{dt} = G(x, t, u), \tag{42}$$

where u is a controllable agricultural action.

An economically relevant stopping boundary can be represented by

$$\Phi(x, t) = 0. \tag{43}$$

Production continues while

$$\Phi(x, t) > 0, \tag{44}$$

and stopping occurs when

$$\Phi(x, t) \leq 0. \tag{45}$$

The original agricultural condition can therefore be interpreted as a particularly simple threshold relation.

13 Statistics

The principal empirical advantage of the framework is observability.

Suppose observations are indexed by t. The basic quantities can be estimated as

$$\widehat{y}_t = \frac{P_t}{s_t} - 1, \tag{46}$$

$$\widehat{d}_t = \frac{F_t}{P_t} - 1, \tag{47}$$

and

$$\widehat{r}_t = \frac{F_t}{s_t} - 1. \tag{48}$$

The empirical stopping gap can then be defined as

$$G_t = 1 + \widehat{y}_t - \frac{1 + \widehat{r}_t}{1 + \widehat{d}_t}. \tag{49}$$

The proposed stopping boundary is

$$G_t = 0. \tag{50}$$

A tolerance-based operational rule could be

$$|G_t| \leq \varepsilon, \tag{51}$$

where ε is a prespecified estimation tolerance.

This converts the theoretical stopping condition into an empirically measurable statistic.

14 AI and machine learning

The stopping condition is particularly amenable to AI/ML because the underlying variables can be estimated from observable agricultural information.

Potential data sources include:

- satellite imagery;
- drone imagery;
- field sensors;
- weather observations;
- soil measurements;
- irrigation records;
- historical yields;
- market arrivals;
- commodity prices;
- storage inventories;
- fertilizer application;
- disease and pest observations.

A supervised learning system could estimate future values of P and F from observed state variables.

Let the state vector be

$$\mathbf{X}_t = (s_t, w_t, n_t, T_t, H_t, M_t, \ldots)^{\mathsf{T}}, \tag{52}$$

where the components may represent seed quantity, water, nutrients, temperature, humidity, market conditions, and other information.

A prediction model may then produce

$$\widehat{P}_{t+h} = f_\theta(\mathbf{X}_t), \tag{53}$$

and

$$\widehat{F}_{t+h} = g_\phi(\mathbf{X}_t). \tag{54}$$

The corresponding predicted stopping gap becomes

$$\widehat{G}_{t+h} = 1 + \frac{\widehat{P}_{t+h}}{s_t} - 1 - \frac{1 + \left(\frac{\widehat{F}_{t+h}}{s_t} - 1\right)}{1 + \left(\frac{\widehat{F}_{t+h}}{\widehat{P}_{t+h}} - 1\right)}. \tag{55}$$

An operational AI system could estimate the probability that the stopping boundary will be reached:

$$\Pr(G_{t+h} \leq \varepsilon \mid \mathbf{X}_t). \tag{56}$$

15 Algorithmic implementation

A simple decision algorithm is:

```
INPUT:
seed observations
plant observations
fruit observations
environmental variables
market variables
tolerance epsilon

ESTIMATE:
y = P/s - 1
d = F/P - 1
r = F/s - 1

COMPUTE:
G = 1 + y - (1 + r)/(1 + d)

IF |G| <= epsilon:
FLAG stopping boundary
ELSE:
CONTINUE production analysis

OPTIONALLY:
use ML forecasts for future P and F
estimate probability of reaching the boundary
incorporate predicted prices and costs
evaluate welfare and competitive effects

OUTPUT:
estimated stopping state
confidence or probability estimate
expected price/revenue consequences
recommended production decision
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The algorithm is intentionally transparent. More sophisticated reinforcement-learning or model-predictive-control systems could subsequently optimize continuation decisions subject to biological, financial, environmental, and market constraints.

16 Psychology and behavioral economics

A stopping rule can also address a behavioral problem: the tendency to equate "more production" with "better performance."

A producer may experience a psychological incentive to continue because:

- previous investment creates a sunk-cost effect;
- high recent prices create extrapolative expectations;
- visible production success creates optimism;
- competitors' behavior creates strategic anxiety;
- stopping may be perceived as failure;
- uncertainty makes immediate continuation psychologically easier than deliberate stopping.

These mechanisms can cause actual behavior to deviate from a mathematically specified stopping boundary.

The framework therefore separates two questions:

$$\text{Can production continue?} \tag{57}$$

and

$$\text{Should production continue?} \tag{58}$$

The first is biological and technological. The second is economic and strategic.

17 Psychiatry and decision stability

Clinical psychiatry is not required to establish the agricultural model, and no psychiatric diagnosis is implied by ordinary agricultural decision errors. Nevertheless, decision science recognizes that cognitive and affective states can influence risk assessment, persistence, impulsivity, and responses to uncertainty.

Accordingly, a production-management system should ideally rely on explicit quantitative thresholds rather than solely on subjective confidence.

The relevant engineering principle is:

$$\text{human judgment} + \text{objective measurement} \longrightarrow \text{more disciplined decision-making.} \tag{59}$$

AI should therefore function as decision support rather than as an unquestioned autonomous authority.

18 Welfare economics

The private producer's optimal stopping decision need not coincide with the social optimum.

Let producer profit be

$$\Pi(Q) = R(Q) - C(Q). \tag{60}$$

Let consumer surplus be $CS(Q)$ and external environmental cost be $E(Q)$. A stylized social welfare function is

$$W(Q) = \Pi(Q) + CS(Q) - E(Q). \tag{61}$$

The private and social stopping points differ whenever

$$\frac{dE}{dQ} \neq 0 \tag{62}$$

or when other externalities or market failures exist.

Agricultural policy therefore has an interest in avoiding both shortages and burdensome surpluses. The broader agricultural-economics literature also documents cases in which observed agricultural markets can simultaneously exhibit underpricing and overproduction relative to estimated efficient equilibria [3].

19 Political economy

Agricultural production is rarely governed by markets alone.

Governments may intervene through:

- minimum support prices;
- procurement;
- subsidies;
- export restrictions;
- import tariffs;
- strategic reserves;
- food-security programs;
- insurance;
- credit policy;
- irrigation and infrastructure investment.

These policies can change the economic consequences of continuation and stopping.

A private producer may rationally continue production if government procurement guarantees a price that would not otherwise be available. Conversely, government intervention may create inventory accumulation or distort production incentives if procurement exceeds sustainable demand.

Thus the stopping rule should ultimately be embedded within the institutional environment rather than treated as an isolated farm-level equation.

20 Geopolitics and international relations

Agriculture also has strategic significance.

Food production affects:

$$\text{food security} \longrightarrow \text{political stability} \longrightarrow \text{national security}. \tag{63}$$

International agricultural trade creates additional transmission channels. A country that produces a large surplus may seek export markets. Multiple countries pursuing simultaneous surplus production can intensify international price competition.

Conversely, a country that stops production too aggressively may become dependent upon imports.

The relevant national decision is therefore not simply

$$\max Q, \tag{64}$$

but rather a multidimensional objective involving production, prices, reserves, resilience, trade exposure, and strategic autonomy.

21 Warfare economics and strategic resilience

Agricultural output has historically been strategically relevant because food is an essential input into population survival and military capability.

A modern national agricultural system therefore has at least two competing objectives:

$$\text{economic efficiency} \tag{65}$$

and

$$\text{strategic resilience}. \tag{66}$$

An optimal stopping rule should consequently distinguish between commercial surplus and strategically necessary reserves.

Let S denote strategic stock. A government may impose

$$S \geq S_{\min}. \tag{67}$$

The economically appropriate stopping point for ordinary commercial production can then differ from the national-security requirement.

This demonstrates why the agricultural stopping problem is inherently multidimensional when considered at national scale.

22 Integrated interdisciplinary framework

The principal causal architecture can be represented as follows.

$$\begin{tikzcd}
\text{Seeds} \arrow[r] & \text{Plants} \arrow[r] & \text{Fruits} \arrow[r] & \text{Market} \arrow[d] \\
& \text{Stopping decision} \arrow[u] & & \text{Price} \arrow[ll]
\end{tikzcd}$$

The agricultural system can therefore be understood as a feedback system:

$$\text{production} \to \text{output} \to \text{market supply} \to \text{price} \to \text{economic incentive} \to \text{production decision}. \tag{68}$$

The original paper provides the compact production-side algebra. The present paper adds the economic feedback interpretation.

23 A unified stopping criterion

A more complete decision system can combine the original stopping gap with economic, biological, and environmental variables.

Define the production stopping gap:

$$G_t = 1 + y_t - \frac{1 + r_t}{1 + d_t}. \tag{69}$$

Define the economic marginal value:

$$M_t = MR_t - MC_t. \tag{70}$$

Define a market-pressure indicator:

$$C_t = -\frac{\Delta p_t}{\Delta Q_t}, \tag{71}$$

whenever the empirical ratio is meaningful.

Define an environmental-pressure indicator:

$$E_t = \frac{\text{marginal environmental cost}}{\text{marginal output}}. \tag{72}$$

A generalized stopping score can then be written as

$$\Psi_t = \alpha G_t + \beta M_t - \gamma C_t - \delta E_t, \tag{73}$$

where the coefficients represent the relative importance assigned to the corresponding dimensions.

This generalized score is an extension, not part of the original paper.

24 Empirical research design

A serious empirical implementation should distinguish prediction from causal inference.

A panel data structure could be written as

$$Y_{it} = \alpha_i + \lambda_t + \beta X_{it} + \varepsilon_{it}, \tag{74}$$

where i indexes farms or regions and t indexes time.

The model can examine whether deviations from the proposed stopping boundary predict subsequent market outcomes.

For example,

$$\Delta p_{i,t+h} = \alpha_i + \lambda_t + \beta G_{it} + \gamma Z_{it} + \varepsilon_{i,t+h}, \tag{75}$$

where Z_{it} contains controls such as weather, demand, inventories, transportation costs, and policy variables.

A finding that $\beta < 0$ would be consistent with the hypothesis that excessive continuation associated with the stopping-gap measure predicts subsequent price pressure. Such a result would require careful identification before being interpreted causally.

25 Falsification and robustness

The framework should be tested rather than merely confirmed.

Potential falsification tests include:

1. testing whether the stopping gap predicts no price movement in markets where supply is perfectly absorbed;
2. testing whether the relationship disappears when demand expands proportionally with supply;
3. testing alternative definitions of market pressure;
4. testing different forecasting horizons;
5. testing out-of-sample rather than only in-sample prediction;
6. comparing linear models with nonlinear machine-learning models;
7. testing whether policy interventions explain apparent stopping effects;
8. testing whether storage and export capacity mediate the relationship.

The objective is not to force empirical data to conform to equation (1), which is algebraically true by construction, but to determine whether the proposed stopping interpretation has predictive and economic content.

26 The fundamental distinction

The central conceptual distinction of this paper is:

$$\boxed{\text{production capacity} \neq \text{optimal production.}} \tag{76}$$

A producer may be capable of producing additional output.

The relevant question is whether doing so creates greater economic value after accounting for:

- production costs;
- market prices;
- competitor responses;
- storage;
- transportation;

- demand;
- perishability;
- environmental costs;
- government policy;
- strategic reserves;
- uncertainty.

The stopping condition provides a compact mathematical signal around which these additional considerations can be organized.

27 Conclusion

The original paper's agricultural framework is exceptionally compact. It begins with the sequential transformation

$$s \longrightarrow P \longrightarrow F \tag{77}$$

and defines three corresponding proportional quantities:

$$y = \frac{P}{s} - 1, \qquad d = \frac{F}{P} - 1, \qquad r = \frac{F}{s} - 1. \tag{78}$$

These imply the fundamental equation

$$\boxed{r = y + d + yd,} \tag{79}$$

the inverse relation

$$\boxed{y = \frac{r-d}{1+d},} \tag{80}$$

the exact algebraic decomposition

$$\boxed{\frac{r-d}{1+d} = r - d(1-d)(1+d^2)(1+r) - \frac{d^5(1+r)}{1+d},} \tag{81}$$

and the stopping condition

$$\boxed{1 + y = \frac{1+r}{1+d}.} \tag{82}$$

The economic significance of the stopping condition arises from its potential operational simplicity. Because its inputs can be constructed from observable agricultural quantities, the condition can be estimated, forecast, monitored, and potentially embedded into AI/ML decision-support systems.

The economic extension developed here is that failing to stop at an economically appropriate point can expose producers to the consequences of excess supply: falling market prices, revenue compression, strategic price competition, and competitor undercutting. This interpretation is compatible with established agricultural-market research concerning oversupply, price pressure, market disequilibrium, and competitive behavior [2, 3, 4].

The strongest version of the proposition is therefore not that agriculture should produce less. It is:

$$\boxed{\text{Agriculture should produce until the economically relevant stopping boundary is reached.}} \tag{83}$$

The distinction between "more" and "optimal" is fundamental. A technologically productive agricultural system can still be economically inefficient if it continues producing output whose marginal social or private value is inadequate.

The next empirical step is therefore clear: construct time-series or panel datasets containing seeds, plants, fruits, prices, costs, inventories, competitors, environmental variables, and policy variables; estimate the stopping boundary; and test whether deviations from that boundary predict subsequent price, revenue, welfare, and competitive outcomes.

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Glossary

Agricultural economics
The study of economic decisions, markets, institutions, resources, production, consumption, and policy associated with agriculture.

Dividend
In the original paper, the proportional transformation from plants to fruits,

$$d = \frac{F}{P} - 1. \tag{84}$$

Yield
In the original paper, the proportional transformation from seeds to plants,

$$y = \frac{P}{s} - 1. \tag{85}$$

Reap
In the original paper, the complete proportional transformation from seeds to fruits,

$$r = \frac{F}{s} - 1. \tag{86}$$

Stopping condition
The condition proposed in the original paper,

$$1 + y = \frac{1 + r}{1 + d}. \tag{87}$$

Optimal stopping
A decision to cease continuation of an activity when continuation no longer satisfies the relevant objective or constraint.

Low-balling
An informal term used in this paper for selling at an unusually low price, particularly under conditions of excess supply or competitive pressure.

Undercutting
Offering a product at a lower price than a competitor in an attempt to obtain or retain market share.

Market pressure
The economic pressure created when available supply, demand, competition, inventories, and institutional conditions influence the price obtainable by producers.

Marginal revenue
The change in revenue generated by an additional unit of output,

$$MR = \frac{dR}{dQ}. \tag{88}$$

Marginal cost

The change in production cost generated by an additional unit of output,

$$MC = \frac{dC}{dQ}. \tag{89}$$

Stopping gap

The deviation from the proposed stopping equality,

$$G_t = 1 + y_t - \frac{1 + r_t}{1 + d_t}. \tag{90}$$

AI/ML

Artificial intelligence and machine learning methods used for prediction, classification, estimation, control, and decision support.

Strategic resilience

The capacity of an agricultural system or nation to maintain essential food-production and food-security capabilities under adverse conditions.

The End